

COS203 DISCRETE STRUCTURES

200 Multiple-Choice Practice Questions

No answers included - use this as a group revision booklet

Prepared from the course material shared in this chat and aligned to the main COS203 Discrete Structures topics.

Unit 1 - Set Theory

1. Which of the following best describes the sentence: "5 is larger than 7"?
 - A. Statement
 - B. Question
 - C. Command
 - D. Open sentence
2. Which of the following best describes the sentence: "Would you like some ice cream?"?
 - A. Question
 - B. Command
 - C. Statement
 - D. Open sentence
3. Which of the following best describes the sentence: " $4 + x = 12$ "?
 - A. Open sentence
 - B. Statement
 - C. Question
 - D. Command
4. Which of the following best describes the sentence: "Mobile numbers in Nigeria have 11 digits"?
 - A. Statement
 - B. Open sentence
 - C. Command
 - D. Question
5. Which of the following best describes the sentence: "Go to the lecture room!"?
 - A. Command
 - B. Question
 - C. Statement
 - D. Open sentence
6. Which of the following best describes the sentence: "Every even number greater than 2 can be expressed as the sum of two primes"?
 - A. Statement
 - B. Command
 - C. Question
 - D. Open sentence
7. Which of the following best describes the sentence: "The product of two numbers"?
 - A. Open sentence
 - B. Statement
 - C. Question
 - D. Command
8. Which of the following best describes the sentence: "12 is a perfect square"?
 - A. Statement
 - B. Open sentence
 - C. Question
 - D. Command
9. Which of the following best describes the sentence: " $x > 5$ "?
 - A. Open sentence
 - B. Statement
 - C. Question
 - D. Command

10. Which of the following best describes the sentence: "The sun is a star"?
- A. Statement
 - B. Open sentence
 - C. Question
 - D. Command
11. Which option correctly evaluates: $\{3\}$ subset of $\{1,2,3\}$?
- A. True
 - B. False
 - C. Meaningless
 - D. Cannot be determined
12. Which option correctly evaluates: 3 in $\{1,2,3\}$?
- A. True
 - B. False
 - C. Meaningless
 - D. Cannot be determined
13. Which option correctly evaluates: 3 subset of $\{1,2,3\}$?
- A. Meaningless
 - B. True
 - C. False
 - D. Cannot be determined
14. Which option correctly evaluates: empty set subset of $\{2,4,6\}$?
- A. True
 - B. False
 - C. Meaningless
 - D. Cannot be determined
15. Which option correctly evaluates: empty set in $\{2,4,6\}$?
- A. False
 - B. True
 - C. Meaningless
 - D. Cannot be determined
16. Which option correctly evaluates: $\{2,4\}$ proper subset of $\{2,4,6\}$?
- A. True
 - B. False
 - C. Meaningless
 - D. Cannot be determined
17. Which option correctly evaluates: $\{2,4,6\}$ proper subset of $\{2,4,6\}$?
- A. False
 - B. True
 - C. Meaningless
 - D. Cannot be determined
18. Which option correctly evaluates: 7 not in $\{1,3,5,7\}$?
- A. False
 - B. True
 - C. Meaningless
 - D. Cannot be determined

19. Which option correctly evaluates: $\{7\}$ in $\{1,3,\{7\}\}$?
- A. True
 - B. False
 - C. Meaningless
 - D. Cannot be determined
20. Which option correctly evaluates: $\{7\}$ subset of $\{1,3,\{7\}\}$?
- A. False
 - B. True
 - C. Meaningless
 - D. Cannot be determined
21. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, what is A union B ?
- A. $\{3, 4\}$
 - B. $\{1, 2\}$
 - C. $\{5, 6\}$
 - D. $\{1, 2, 3, 4, 5, 6\}$
22. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, what is A intersection B ?
- A. $\{1, 2\}$
 - B. $\{5, 6\}$
 - C. $\{3, 4\}$
 - D. $\{1, 2, 3, 4, 5, 6\}$
23. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, what is $A \setminus B$?
- A. $\{5, 6\}$
 - B. $\{1, 2\}$
 - C. $\{1, 2, 3, 4, 5, 6\}$
 - D. $\{3, 4\}$
24. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, what is $B \setminus A$?
- A. $\{5, 6\}$
 - B. $\{1, 2, 3, 4, 5, 6\}$
 - C. $\{3, 4\}$
 - D. $\{1, 2\}$
25. If $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5\}$, what is A union B ?
- A. $\{6\}$
 - B. $\{1, 3, 5\}$
 - C. $\{1, 2, 3, 4, 5, 6\}$
 - D. $\{2, 4\}$
26. If $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5\}$, what is A intersection B ?
- A. $\{1, 3, 5\}$
 - B. $\{2, 4\}$
 - C. $\{1, 2, 3, 4, 5, 6\}$
 - D. $\{6\}$
27. If $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5\}$, what is $A \setminus B$?
- A. $\{6\}$
 - B. $\{1, 2, 3, 4, 5, 6\}$
 - C. $\{2, 4\}$
 - D. $\{1, 3, 5\}$

28. If $A = \{2, 4, 6\}$ and $B = \{1, 2, 3, 4, 5\}$, what is $B \setminus A$?
- A. $\{1, 2, 3, 4, 5, 6\}$
 - B. $\{2, 4\}$
 - C. $\{6\}$
 - D. $\{1, 3, 5\}$
29. If $A = \{1, 3, 5, 7\}$ and $B = \{3, 5, 8, 9\}$, what is $A \cup B$?
- A. $\{8, 9\}$
 - B. $\{1, 3, 5, 7, 8, 9\}$
 - C. $\{3, 5\}$
 - D. $\{1, 7\}$
30. If $A = \{1, 3, 5, 7\}$ and $B = \{3, 5, 8, 9\}$, what is $A \cap B$?
- A. $\{3, 5\}$
 - B. $\{1, 3, 5, 7, 8, 9\}$
 - C. $\{1, 7\}$
 - D. $\{8, 9\}$
31. If $A = \{1, 3, 5, 7\}$ and $B = \{3, 5, 8, 9\}$, what is $A \setminus B$?
- A. $\{1, 3, 5, 7, 8, 9\}$
 - B. $\{3, 5\}$
 - C. $\{8, 9\}$
 - D. $\{1, 7\}$
32. If $A = \{1, 3, 5, 7\}$ and $B = \{3, 5, 8, 9\}$, what is $B \setminus A$?
- A. $\{3, 5\}$
 - B. $\{1, 7\}$
 - C. $\{8, 9\}$
 - D. $\{1, 3, 5, 7, 8, 9\}$
33. If $A = \{1, 2, 4, 5, 6\}$ and $B = \{2, 3, 4\}$, what is $A \cup B$?
- A. $\{1, 2, 3, 4, 5, 6\}$
 - B. $\{2, 4\}$
 - C. $\{1, 5, 6\}$
 - D. $\{3\}$
34. If $A = \{1, 2, 4, 5, 6\}$ and $B = \{2, 3, 4\}$, what is $A \cap B$?
- A. $\{1, 2, 3, 4, 5, 6\}$
 - B. $\{1, 5, 6\}$
 - C. $\{3\}$
 - D. $\{2, 4\}$
35. If $A = \{1, 2, 4, 5, 6\}$ and $B = \{2, 3, 4\}$, what is $A \setminus B$?
- A. $\{2, 4\}$
 - B. $\{3\}$
 - C. $\{1, 5, 6\}$
 - D. $\{1, 2, 3, 4, 5, 6\}$
36. If $A = \{1, 2, 4, 5, 6\}$ and $B = \{2, 3, 4\}$, what is $B \setminus A$?
- A. $\{1, 5, 6\}$
 - B. $\{3\}$
 - C. $\{1, 2, 3, 4, 5, 6\}$
 - D. $\{2, 4\}$

37. What is the number of elements in the power set of $\{a, b\}$?
- A. 4
 - B. 2
 - C. 3
 - D. 8
38. What is the number of elements in the power set of $\{1,2,3\}$?
- A. 6
 - B. 4
 - C. 9
 - D. 8
39. What is the number of elements in the power set of $\{x, y, z, w\}$?
- A. 12
 - B. 32
 - C. 16
 - D. 8
40. What is the number of elements in the power set of empty set ?
- A. 4
 - B. 1
 - C. 0
 - D. 2
41. What is the number of elements in the power set of $\{p, q, r, s, t\}$?
- A. 32
 - B. 16
 - C. 10
 - D. 64
42. Which option best completes the statement: $\{1,2,3\}$ and $\{4,5,6\}$?
- A. Disjoint sets
 - B. Equal sets
 - C. Overlapping sets
 - D. Universal sets
43. Which option best completes the statement: $\{1,2,3\}$ and $\{3,4,5\}$?
- A. Overlapping sets
 - B. Disjoint sets
 - C. Equal sets
 - D. Empty sets
44. Which option best completes the statement: A relation that is reflexive, symmetric and transitive is called?
- A. An equivalence relation
 - B. A function
 - C. A partial order
 - D. A binary operation

Unit 2 - Proofs and Induction

45. Which proof method starts by assuming the premise is true and then logically derives the conclusion?
- A. Direct proof
 - B. Proof by contradiction
 - C. Proof by contrapositive
 - D. Proof by exhaustion

46. In induction, the first step is called the
- A. Base case
 - B. Contrapositive step
 - C. Induction fallacy
 - D. Pigeonhole step
47. In induction, assuming the statement is true for $n = k$ is called the
- A. Induction hypothesis
 - B. Base case
 - C. Conclusion
 - D. Counterexample
48. The step that proves the statement for $n = k + 1$ is called the
- A. Inductive step
 - B. Base case
 - C. Contradiction step
 - D. Truth-table step
49. If a statement $P \rightarrow Q$ is proved by showing $\text{not } Q \rightarrow \text{not } P$, the method is
- A. Contrapositive
 - B. Direct proof
 - C. Induction
 - D. Case analysis
50. If a proof assumes P and $\text{not } Q$ and reaches a contradiction, the method is
- A. Contradiction
 - B. Direct proof
 - C. Contrapositive
 - D. Enumeration
51. The definition of an even integer is
- A. $2k$
 - B. $2k+1$
 - C. k^2
 - D. $k-1$
52. The definition of an odd integer is
- A. $2k+1$
 - B. $2k$
 - C. k^2+1
 - D. $k+2$
53. If $n = 2k$, then n^2 can be written as
- A. $4k^2$
 - B. $2k^2$
 - C. $2k+1$
 - D. k^2+2
54. If $n = 2k+1$, then $(2k+1)^2$ expands to
- A. $4k^2 + 4k + 1$
 - B. $4k^2 + 2k + 1$
 - C. $2k^2 + 1$
 - D. $4k + 1$

55. The expression $4k^2$ can be rewritten as
- A. $2(2k^2)$
 - B. $2(k^2)$
 - C. $4(2k)$
 - D. $k(4k)$
56. If n is even, which form should be used in a proof?
- A. $n = 2k$
 - B. $n = 2k+1$
 - C. $n = k^2$
 - D. $n = 2k-1$
57. If n is odd, which form should be used in a proof?
- A. $n = 2k+1$
 - B. $n = 2k$
 - C. $n = k^2$
 - D. $n = 2k^2$
58. To prove that a number is a multiple of 3, a common strategy is to write it as
- A. $3k$
 - B. $3k+1$
 - C. k^3
 - D. $2k+1$
59. If n is an integer, the expression $n(n+1)$ is always
- A. Even
 - B. Odd
 - C. Prime
 - D. Negative
60. If n is even, then $n+6$ is even because
- A. $2k+6 = 2(k+3)$
 - B. $2k+6 = 3(k+2)$
 - C. $2k+6 = k+6$
 - D. $2k+6 = 2k^2$
61. The base case for a standard induction proof is usually checked at
- A. $n = 1$
 - B. $n = 0$
 - C. $n = 2$
 - D. $n = k$
62. The statement to be proved in the induction hypothesis is assumed true for
- A. $n = k$
 - B. $n = k+1$
 - C. $n = 1$ only
 - D. $n = 0$
63. In the inductive step, you must prove the statement for
- A. $n = k+1$
 - B. $n = k-1$
 - C. $n = 2k$
 - D. $n = 0$

- 64.** A proof by induction is most suitable for statements about
- A. Positive integers
 - B. Complex numbers only
 - C. Matrices only
 - D. Graphs only
- 65.** The induction step normally uses the
- A. Induction hypothesis
 - B. Base case
 - C. Truth table
 - D. Complement law
- 66.** If $P(1)$ is true and $P(k) \rightarrow P(k+1)$, then by induction the statement is true for
- A. All positive integers
 - B. Only $k = 1$
 - C. Only even integers
 - D. Only prime numbers
- 67.** A common exam pattern for induction is to prove a formula for
- A. A sum or sequence
 - B. A graph coloring only
 - C. A matrix determinant only
 - D. A truth table only
- 68.** To prove $P \rightarrow Q$ by contrapositive, you prove
- A. $\text{not } Q \rightarrow \text{not } P$
 - B. $Q \rightarrow P$
 - C. P and Q
 - D. $P \leftrightarrow Q$
- 69.** A contradiction proof assumes
- A. P and $\text{not } Q$
 - B. P only
 - C. Q only
 - D. $\text{not } P$ and $\text{not } Q$
- 70.** If a proof ends by showing a statement cannot be both true and false, the proof is by
- A. Contradiction
 - B. Induction
 - C. Direct proof
 - D. Enumeration
- 71.** The statement 'If n is even, then n^2 is even' can be proved by
- A. Direct proof or contrapositive
 - B. Only contradiction
 - C. Only tree diagram
 - D. Only pigeonhole principle
- 72.** A counterexample is used to
- A. Disprove a universal statement
 - B. Prove a theorem by induction
 - C. Build a power set
 - D. Create a truth table

Unit 3 - Logic

- 73.** A declarative sentence that is either true or false is called a
- A. Statement
 - B. Command
 - C. Question
 - D. Argument
- 74.** A statement that cannot be divided into smaller statements is
- A. Atomic
 - B. Molecular
 - C. Compound
 - D. Negated
- 75.** A statement formed by combining smaller statements is
- A. Molecular
 - B. Atomic
 - C. Empty
 - D. Universal
- 76.** The connective 'and' is represented by
- A. AND
 - B. OR
 - C. not
 - D. implies
- 77.** The connective 'or' is represented by
- A. OR
 - B. AND
 - C. iff
 - D. not
- 78.** The connective 'not' is represented by
- A. not
 - B. AND
 - C. OR
 - D. implies
- 79.** The connective 'if and only if' is represented by
- A. iff
 - B. implies
 - C. OR
 - D. AND
- 80.** The statement $P \text{ AND } Q$ is true when
- A. Both P and Q are true
 - B. Either P or Q is true
 - C. P is false
 - D. Q is false
- 81.** If P is true and Q is false, then $P \text{ OR } Q$ is
- A. True
 - B. False
 - C. Undefined
 - D. Both true and false

- 82.** If P is true and Q is false, then P AND Q is
- A. False
 - B. True
 - C. Undefined
 - D. Sometimes true
- 83.** If P is false, then not P is
- A. True
 - B. False
 - C. Undefined
 - D. Both
- 84.** If P and Q are both true, then P iff Q is
- A. True
 - B. False
 - C. Undefined
 - D. A contradiction
- 85.** If P is false and Q is true, then P implies Q is
- A. True
 - B. False
 - C. Undefined
 - D. Both
- 86.** If P is true and Q is false, then P implies Q is
- A. False
 - B. True
 - C. Undefined
 - D. A tautology
- 87.** A statement that is always true is called a
- A. Tautology
 - B. Contradiction
 - C. Variable
 - D. Predicate
- 88.** A statement that is always false is called a
- A. Contradiction
 - B. Tautology
 - C. Predicate
 - D. Quantifier
- 89.** The negation of 'for all x, P(x)' is
- A. There exists x such that not P(x)
 - B. For all x, not P(x)
 - C. P(x) is true
 - D. P(x) is false
- 90.** The negation of 'there exists x such that P(x)' is
- A. For all x, not P(x)
 - B. There exists x such that not P(x)
 - C. For all x, P(x)
 - D. P(x) is always true

- 91.** A statement involving 'for all' is a
- A. Universal quantifier
 - B. Existential quantifier
 - C. Boolean operator
 - D. Truth value
- 92.** A statement involving 'there exists' is a
- A. Existential quantifier
 - B. Universal quantifier
 - C. Binary relation
 - D. Set operation
- 93.** If every student passed the test, this is an example of
- A. Universal quantification
 - B. Existential quantification
 - C. Contradiction
 - D. Negation
- 94.** If at least one student passed the test, this is an example of
- A. Existential quantification
 - B. Universal quantification
 - C. Commutativity
 - D. Association
- 95.** Two propositions are logically equivalent if they have
- A. The same truth values in all cases
 - B. Different truth values in all cases
 - C. No truth values
 - D. Only one truth value
- 96.** De Morgan's law for negating a conjunction states that $\text{not}(P \text{ AND } Q)$ equals
- A. $\text{not } P \text{ OR } \text{not } Q$
 - B. $\text{not } P \text{ AND } \text{not } Q$
 - C. $P \text{ OR } Q$
 - D. $P \text{ AND } Q$
- 97.** De Morgan's law for negating a disjunction states that $\text{not}(P \text{ OR } Q)$ equals
- A. $\text{not } P \text{ AND } \text{not } Q$
 - B. $\text{not } P \text{ OR } \text{not } Q$
 - C. $P \text{ OR } Q$
 - D. $P \text{ implies } Q$
- 98.** The statement ' $P \text{ implies } Q$ ' is equivalent to
- A. $\text{not } P \text{ OR } Q$
 - B. $P \text{ AND } Q$
 - C. $\text{not } Q \text{ OR } P$
 - D. $P \text{ iff } Q$
- 99.** The statement ' $P \text{ iff } Q$ ' is true when
- A. P and Q have the same truth value
 - B. P is true and Q is false
 - C. P is false and Q is true
 - D. P is false and Q is false

100. A conditional statement is false only when

- A. P is true and Q is false
- B. P is false and Q is true
- C. Both are true
- D. Both are false

Module 2 - Boolean Algebra and Lattices

101. The idempotent law for union is

- A. $A \cup A = A$
- B. $A \cap A = \text{empty set}$
- C. $A \cup \text{empty set} = U$
- D. $A \cap U = \text{empty set}$

102. The idempotent law for intersection is

- A. $A \cap A = A$
- B. $A \cup A = \text{empty set}$
- C. $A \cap U = U$
- D. $A \cup \text{empty set} = A$

103. The complement law for sets states that $A \cup A^c$ equals

- A. U
- B. empty set
- C. A
- D. A^c

104. The complement law for sets states that $A \cap A^c$ equals

- A. empty set
- B. U
- C. A
- D. A^c

105. De Morgan's law for sets says $(A \cup B)^c$ equals

- A. $A^c \cap B^c$
- B. $A^c \cup B^c$
- C. $A \cap B$
- D. $A \cup B$

106. De Morgan's law for sets says $(A \cap B)^c$ equals

- A. $A^c \cup B^c$
- B. $A^c \cap B^c$
- C. $A \cup B$
- D. $A \cap B$

107. The dual of $A \cup \text{empty set} = A$ is

- A. $A \cap U = A$
- B. $A \cup U = A$
- C. $A \cap \text{empty set} = A$
- D. $A \cup A = \text{empty set}$

108. The dual of $A \cap U = A$ is

- A. $A \cup \text{empty set} = A$
- B. $A \cap \text{empty set} = A$
- C. $A \cup U = U$
- D. $A \cap A = A$

- 109.** The Boolean operator corresponding to logical AND is
- A. AND
 - B. OR
 - C. not
 - D. iff
- 110.** The Boolean operator corresponding to logical OR is
- A. OR
 - B. AND
 - C. not
 - D. xor
- 111.** The Boolean operator corresponding to logical NOT is
- A. not
 - B. AND
 - C. OR
 - D. implies
- 112.** If $A = 1$ and $B = 0$, then $A \text{ AND } B$ is
- A. 0
 - B. 1
 - C. A
 - D. Undefined
- 113.** If $A = 1$ and $B = 0$, then $A \text{ OR } B$ is
- A. 1
 - B. 0
 - C. A
 - D. Undefined
- 114.** A Boolean expression with all variables complemented is often simplified using
- A. De Morgan's laws
 - B. Pigeonhole principle
 - C. Induction
 - D. Hamiltonian paths
- 115.** The principle of duality means that if an identity is true, then its
- A. Dual is also true
 - B. Complement is false
 - C. Matrix inverse is true
 - D. Graph is planar
- 116.** In a lattice, the operations are usually called
- A. Meet and join
 - B. Union and complement
 - C. Path and cycle
 - D. Sum and product
- 117.** A lattice is a partially ordered set in which every pair has
- A. A greatest lower bound and least upper bound
 - B. Only a greatest element
 - C. Only a least element
 - D. No comparable elements

- 118.** The meet of two elements is usually written as
- A. AND
 - B. OR
 - C. xor
 - D. implies
- 119.** The join of two elements is usually written as
- A. OR
 - B. AND
 - C. not
 - D. otimes
- 120.** A bounded lattice has
- A. A greatest and least element
 - B. Only a greatest element
 - C. Only a least element
 - D. No meet operation
- 121.** A distributive lattice satisfies
- A. Distributive laws
 - B. Only commutative law
 - C. Only associative law
 - D. Only complement law
- 122.** In Boolean algebra, every element has a
- A. Complement
 - B. Derivative
 - C. Inverse matrix
 - D. Successor
- 123.** A Boolean algebra as a lattice can be represented by
- A. A poset with meet and join
 - B. A graph with edges
 - C. A matrix equation
 - D. A probability space
- 124.** The complement of the complement of A is
- A. A
 - B. A^c
 - C. empty set
 - D. U

Module 2 - Graph Theory

- 125.** A graph consists of
- A. Vertices and edges
 - B. Tables and rows
 - C. Sets and complements
 - D. Matrices and determinants
- 126.** An edge with a direction is called a
- A. Directed edge
 - B. Loop
 - C. Tree
 - D. Complement

- 127.** A graph with no direction on edges is called
- A. Undirected graph
 - B. Weighted graph
 - C. Complete graph
 - D. Planar graph
- 128.** Two vertices connected by an edge are said to be
- A. Adjacent
 - B. Equivalent
 - C. Complementary
 - D. Parallel
- 129.** The degree of a vertex is the number of
- A. Edges incident on it
 - B. Vertices in the graph
 - C. Faces in the graph
 - D. Paths in the graph
- 130.** The handshaking theorem states that the sum of degrees of all vertices equals
- A. Twice the number of edges
 - B. The number of vertices
 - C. The number of paths
 - D. The number of components
- 131.** If the degree sequence is (4,4,3,3,3,2,1), the number of vertices is
- A. 7
 - B. 10
 - C. 14
 - D. 20
- 132.** If the sum of the degrees of a graph is 20, then the number of edges is
- A. 10
 - B. 20
 - C. 5
 - D. 40
- 133.** A path in a graph is
- A. A sequence of vertices joined by edges
 - B. A set of complements
 - C. A matrix row
 - D. A Boolean law
- 134.** A cycle is a path that
- A. Starts and ends at the same vertex
 - B. Has no edges
 - C. Has only two vertices
 - D. Has no direction
- 135.** An Euler circuit uses every edge exactly
- A. Once
 - B. Twice
 - C. Zero times
 - D. Three times

- 136.** An Euler path is a trail that uses every edge exactly
- A. Once
 - B. Twice
 - C. Three times
 - D. Zero times
- 137.** A tree is a connected graph with no
- A. Cycles
 - B. Vertices
 - C. Edges
 - D. Paths
- 138.** A complete graph on n vertices is denoted by
- A. K_n
 - B. P_n
 - C. C_n
 - D. T_n
- 139.** A bipartite graph can be partitioned into
- A. Two disjoint vertex sets
 - B. Three cycles
 - C. One edge and one loop
 - D. Equal edges and vertices
- 140.** A planar graph can be drawn
- A. Without edge crossings
 - B. Only with loops
 - C. Only with weights
 - D. Only in 3D
- 141.** The adjacency matrix of a graph records
- A. Which vertices are connected
 - B. The determinant of the graph
 - C. The truth value of statements
 - D. The sum of all weights only
- 142.** A graph representation using rows and columns is called
- A. Adjacency matrix
 - B. Power set
 - C. Truth table
 - D. Incidence theorem
- 143.** In an undirected simple graph, the adjacency matrix is
- A. Symmetric
 - B. Skew-symmetric
 - C. Diagonal only
 - D. Always zero
- 144.** A loop at a vertex contributes
- A. 2 to the degree in an undirected graph
 - B. 1 to the degree in all graphs
 - C. 0 to the degree
 - D. 3 to the degree

- 145.** The complement of a graph changes
- A. Edges to non-edges
 - B. Vertices to matrices
 - C. Paths to cycles
 - D. Trees to lattices
- 146.** A graph with every vertex having the same degree is called
- A. Regular
 - B. Complete
 - C. Bipartite
 - D. Planar
- 147.** A graph with exactly two vertices of odd degree must have
- A. An even number of odd-degree vertices overall
 - B. No edges
 - C. Only loops
 - D. A complete matrix
- 148.** A connected acyclic graph is called a
- A. Tree
 - B. Cycle
 - C. Circuit
 - D. Multigraph

Module 3 - Matrices and Determinants

- 149.** A matrix with 2 rows and 3 columns is called
- A. A 2 x 3 matrix
 - B. A 3 x 2 matrix
 - C. A square matrix
 - D. A diagonal matrix
- 150.** The transpose of a matrix is obtained by
- A. Swapping rows and columns
 - B. Changing every entry to zero
 - C. Adding a row
 - D. Taking the determinant
- 151.** If A is a square matrix and $\det(A) = 0$, then A is
- A. Singular
 - B. Invertible
 - C. Diagonal
 - D. Symmetric
- 152.** If $\det(A) \neq 0$, then A is
- A. Nonsingular
 - B. Singular
 - C. Triangular only
 - D. Zero matrix
- 153.** The determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is
- A. $ad - bc$
 - B. $ab - cd$
 - C. $ac - bd$
 - D. $a+b+c+d$

- 154.** A matrix multiplied by the identity matrix remains
- A. The same matrix
 - B. The zero matrix
 - C. A transpose
 - D. A determinant
- 155.** The system $x + y = 2$ and $x - y = 0$ has solution
- A. (1,1)
 - B. (2,0)
 - C. (0,2)
 - D. (1,2)
- 156.** The determinant of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is
- A. -2
 - B. 2
 - C. 4
 - D. 0
- 157.** The determinant of $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ is
- A. 10
 - B. 7
 - C. 0
 - D. 5
- 158.** A 3×3 matrix has how many entries?
- A. 9
 - B. 6
 - C. 3
 - D. 12
- 159.** If two rows of a determinant are identical, the determinant is
- A. 0
 - B. 1
 - C. Undefined
 - D. Negative
- 160.** If a matrix is upper triangular, its determinant equals the
- A. Product of diagonal entries
 - B. Sum of all entries
 - C. Number of rows
 - D. Determinant of its transpose only
- 161.** The inverse of a matrix exists only when the matrix is
- A. Nonsingular
 - B. Singular
 - C. Diagonal only
 - D. Symmetric only
- 162.** Gaussian elimination is used mainly to
- A. Solve linear systems
 - B. Find graph cycles
 - C. Build truth tables
 - D. Generate power sets

- 163.** The product AB is defined when
- A. The number of columns of A equals the number of rows of B
 - B. A and B are both square
 - C. A and B have same determinant
 - D. A and B are diagonal
- 164.** If A is 2×3 and B is 3×4 , then AB is
- A. 2×4
 - B. 3×3
 - C. 4×2
 - D. 2×3
- 165.** The identity matrix is usually denoted by
- A. I
 - B. 0
 - C. J
 - D. T
- 166.** A diagonal matrix has nonzero entries only on the
- A. Main diagonal
 - B. Last row
 - C. First column
 - D. Anti-diagonal only
- 167.** The trace of a square matrix is the sum of its
- A. Diagonal entries
 - B. Off-diagonal entries
 - C. Rows
 - D. Columns
- 168.** If A is equal to its transpose, A is
- A. Symmetric
 - B. Singular
 - C. Triangular
 - D. Regular

Module 3 - Applications to Counting

- 169.** If task A can be done in 3 ways and task B in 4 ways, then performing both tasks in sequence can be done in
- A. 12 ways
 - B. 7 ways
 - C. 1 way
 - D. 24 ways
- 170.** The sum rule is used when the choices are
- A. Mutually exclusive
 - B. All simultaneous
 - C. Dependent only
 - D. Matrix-based
- 171.** A permutation is an arrangement where
- A. Order matters
 - B. Order does not matter
 - C. Only sets are used
 - D. No repetition is allowed always

- 172.** A combination is a selection where
- A. Order does not matter
 - B. Order matters
 - C. Only one item is chosen
 - D. The same as permutation
- 173.** The number of ways to arrange 4 distinct objects is
- A. 24
 - B. 12
 - C. 8
 - D. 16
- 174.** The number of ways to choose 2 objects from 5 is
- A. 10
 - B. 20
 - C. 5
 - D. 15
- 175.** The pigeonhole principle says that if more objects than boxes are placed, then at least one box gets
- A. At least two objects
 - B. No objects
 - C. Exactly one object
 - D. Exactly one box
- 176.** The inclusion-exclusion principle is mainly used to
- A. Count without double counting
 - B. Build matrices
 - C. Prove symmetry
 - D. Find determinants
- 177.** If there are 10 students and each chooses one of 3 clubs, the product rule gives
- A. 30 possible student-club assignments
 - B. 13
 - C. 17
 - D. 7
- 178.** The number of permutations of 3 letters chosen from 5 distinct letters is
- A. 60
 - B. 10
 - C. 15
 - D. 125
- 179.** A tree diagram is useful in counting when
- A. Choices occur step by step
 - B. A determinant is needed
 - C. The graph is complete
 - D. The set is empty
- 180.** Pascal's triangle is closely related to
- A. Binomial coefficients
 - B. Graph colors
 - C. Matrix inverses
 - D. Truth tables

181. In nPr , the P stands for

- A. Permutation
- B. Power
- C. Probability
- D. Partition

182. In nCr , the C stands for

- A. Combination
- B. Cycle
- C. Cardinality
- D. Commutativity

183. The formula nCr equals $n!$ divided by

- A. $r!(n-r)!$
- B. $r!n!$
- C. $(n+r)!$
- D. $n!r!$

184. If 6 people shake hands once each with every other person, the handshaking count is based on

- A. Pairs
- B. Permutations
- C. Matrices
- D. Power sets

Module 3 - Discrete Probability Generating Function

185. A sample space is

- A. The set of all possible outcomes
- B. The set of all events only
- C. A graph representation
- D. A matrix of outcomes

186. An event is

- A. A subset of the sample space
- B. A determinant
- C. A power set of matrices
- D. A Boolean law

187. If $P(A) = 0.2$ and $P(B|A) = 0.5$, then conditional probability is about

- A. Probability given A occurred
- B. Combination count
- C. Graph traversal
- D. Matrix rank

188. Two events are independent if

- A. $P(A \text{ and } B) = P(A)P(B)$
- B. $P(A)=P(B)$
- C. A and B are equal
- D. A is a subset of B

189. The probability generating function of X is often written as

- A. $G_X(z)$
- B. $\det(X)$
- C. $P(X)$
- D. $V(X)$

- 190.** For a binomial random variable $X \sim B(n, p)$, the PGF is
- A. $(pz + q)^n$
 - B. $(p+q)^n$
 - C. $e^{\lambda z}$
 - D. $1/(1-z)$
- 191.** For a geometric random variable with parameter p , the PGF is
- A. $p/(1-qz)$
 - B. $(pz+q)^n$
 - C. $e^{\lambda(z-1)}$
 - D. $1/z$
- 192.** For a Poisson random variable with parameter λ , the PGF is
- A. $e^{\lambda(z-1)}$
 - B. $(pz+q)^n$
 - C. $p/(1-qz)$
 - D. λz
- 193.** $G_X(1)$ for any valid PGF is
- A. 1
 - B. 0
 - C. The mean
 - D. Negative
- 194.** $G_X(0)$ equals
- A. $P(X=0)$
 - B. $P(X=1)$
 - C. 1
 - D. The variance
- 195.** The mean of a distribution can be found from a PGF using
- A. Derivatives
 - B. Subtraction
 - C. Graph coloring
 - D. Matrix inversion
- 196.** The variance of a distribution is also linked to
- A. Derivatives of the PGF
 - B. Set union
 - C. Truth tables
 - D. Euler circuits
- 197.** A discrete random variable takes
- A. Countable values
 - B. Only one value
 - C. No values
 - D. Continuous values only
- 198.** A binomial experiment requires
- A. Fixed number of trials and two outcomes
 - B. Infinite outcomes
 - C. A graph
 - D. A matrix

- 199.** A geometric distribution is associated with
- A. Repeated independent trials until first success
 - B. Determinants
 - C. Trees only
 - D. Boolean complements
- 200.** A Poisson distribution is commonly used for
- A. Counts of events in a fixed interval
 - B. Exact matrix rank
 - C. Graph coloring
 - D. Power sets